A Sound Single Convex Program for Powered-Descent Guidance of a Gimbaled Lander

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Lossless convexification solves three-degree-of-freedom powered-descent guidance as one convex program, but it treats the thrust direction as a free control. On a lander steered by a single gimbaled engine, the thrust direction is tied to a body that the same gimbal must rotate. We first explain why no convex program of the lossless type can be exact for such a vehicle: the set of initial states from which a landing exists is not convex. We then show what remains possible. The center of percussion of the vehicle is not accelerated by the lateral force of the gimbal, so its path determines the attitude, the gimbal angle, and the throttle algebraically. Planning that path as a quintic B-spline, with the actuator limits imposed as convex restrictions on its Bézier control points, gives one second-order cone program with 63 unknown coefficients. We prove that every feasible point of this program is a trajectory of the rigid-body model that meets the throttle, gimbal, and tilt limits at every instant. The program gives up 0.1% of fuel at equal flight time and at most 0.4 s of minimum flight time. In closed-loop simulation of an 80-kg lander with each planner's computation time counted, it replans ten times per second and lands within 1.1 cm in all 40 in-air descents, whereas six-degree-of-freedom successive convexification lands within 18.3 cm.

I. Nomenclature

a	= axial thrust per unit mass, m/s^2	T_a, $T_\perp$	= axial and lateral parts of the thrust, N
b	= unit vector along the body axis	$T_{\min}$, $T_{\max}$	= thrust limits, N
e_z	= unit vector pointing up	t_f	= time of flight, s
f	= $\|u\|$, m/s^2	u	= $\ddot{y} + g_0 e_z$, m/s^2
g_0	= gravitational acceleration, m/s^2	U, U_1, U_2	= control points of $u, \dot{u}, \ddot{u}$
h	= length of one spline span, s	y	= position of the center of percussion, m
J_t	= transverse moment of inertia, kg m^2	α	= mass flow per unit thrust, kg/(N s)
L	= distance from pivot to center of mass, m	$\delta_{\max}$	= gimbal-angle limit, rad
m, m_0	= mass and initial mass, kg	ε	= center-of-percussion offset $J_t/(mL)$, m
p	= vertical component of U, m/s^2	$\vartheta_{\max}$	= tilt limit, rad
r, v	= position and velocity of the center of mass	Π_b	= projector $I - bb^{\mathsf{T}}$
S	= number of spline spans	σ	= bound on thrust per unit mass, m/s^2
T	= thrust vector, N	τ	= thrust per unit mass, m/s^2

II. Introduction

Powered-descent guidance must find a thrust history that brings a lander from its current state to rest at a target without violating the limits of its engine. The lower thrust limit makes the admissible thrust set an annulus, which

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Table 1 Vehicle and planning parameters

Quantity	Value	Quantity	Value
Wet mass, dry mass	80 kg, 55.9 kg	Thrust range, physical	400 to 1000 N
Transverse inertia J_t (wet)	63.6 kg m^2	Thrust range, planning	440 to 920 N
Pivot to center of mass L (wet)	0.589 m	Gimbal limit, physical	±15 deg per axis, 100 deg/s
Center-of-percussion offset ε	1.35 m	Gimbal limit, planning $\delta_{\max}$	10 deg cone
Specific impulse	180 s	Tilt limit $\vartheta_{\max}$	20 deg

is not convex. Lossless convexification (LCvx) replaces the annulus by a convex cone in a lifted space and proves that the solution of the relaxed second-order cone program (SOCP) is feasible for the original problem [1–3]. The result is exact, it needs no initial guess, and it has flown [4]. It is, however, a three-degree-of-freedom (3-DOF) result: the thrust direction is a control that may change instantly.

A vehicle steered by thrust-vector control does not have that freedom. Its thrust acts along an engine that deflects only a few degrees from the body, and the body is turned by that same deflection. Six-degree-of-freedom (6-DOF) guidance therefore uses successive convexification (SCvx): the dynamics are linearized about a trial trajectory, a convex problem is solved inside a trust region, and the process is repeated [5–7]. SCvx handles general constraints, but it needs an initial trajectory, penalty weights, and an iteration budget, and its iterates satisfy the true dynamics only at convergence.

This paper asks what a single convex program can do for a gimbaled lander, and gives two answers. The first is negative. A program is *sound* if it is feasible only when a landing exists and *complete* if it is feasible whenever one exists; LCvx is both. We explain in Sec. 4 why no convex program that takes the initial position and velocity as LCvx does can be both for a vehicle whose thrust direction turns at a finite rate. The second answer is positive, and it is the main contribution: soundness alone is attainable with one SOCP. The construction rests on a classical fact. The center of percussion of the vehicle is not accelerated by the lateral force of the gimbal, so in that point's coordinates the gimbaled rigid body is differentially flat [8]. The same point is the flat output of the planar vertical-takeoff aircraft [9]. Planning in flat outputs is standard for multirotors [10] and has been proposed for planetary landing [11]; imposing constraints on spline control points so that they hold between grid points is also established [12], and B-splines have been used inside sequential convex programming for powered descent [13]. We did not find these ingredients combined as they are here: the center-of-percussion output of the three-dimensional gimbaled body, with the minimum-throttle and gimbal limits imposed as convex restrictions, in one SOCP that is proved to return only flyable trajectories.

The contributions are 1) a single SOCP for the gimbaled lander with depleting mass (Sec. 5); 2) a proof that each of its feasible points satisfies the dynamics exactly and the actuator limits at every instant, and a proof that it cannot also be complete (Sec. 6); and 3) a simulation study against 6-DOF SCvx in which the time each planner takes to compute is charged to it (Sec. 8).

III. Vehicle Model and the Center of Percussion

The vehicle is a rigid axisymmetric body with mass m, transverse inertia J_t, and unit body axis b (Fig. 1). The engine pivots on the body axis a distance L below the center of mass and delivers the thrust $T = T_a b + T_\perp$, where $T_\perp \perp b$ is the part produced by deflecting the gimbal. A reaction-control system holds the roll rate at zero, so the angular velocity Ω is perpendicular to b. With r the position of the center of mass and α the mass flow per unit thrust, the equations of motion are

$$m\ddot{r} = T - mg_0 e_z, \qquad J_t\dot{\Omega} = (-Lb)\times T, \qquad \dot{b} = \Omega \times b, \qquad \dot{m} = -\alpha\|T\|. \tag{1}$$

The actuator limits are $T_{\min} \le \|T\| \le T_{\max}$, a gimbal angle $\mu \le \delta_{\max}$ with $\tan\mu = \|T_\perp\|/T_a$, and a tilt $\angle(b, e_z) \le \vartheta_{\max} < \pi/2$. Table 1 lists the values used in this paper.

Deflecting the gimbal produces the torque that turns the body, but it also pushes the center of mass sideways. One point of the body is not accelerated by that push. Because $\Omega \perp b$, the lateral part of $\ddot{b}$ is $\Pi_b\ddot{b} = \dot{\Omega}\times b$ with $\Pi_b = I - bb^{\mathsf{T}}$, and the second of Eqs. (1) gives $J_t\,\dot{\Omega}\times b = -LT_\perp$. Hence

$$T_\perp = -m\varepsilon\,\Pi_b\ddot{b}, \qquad \varepsilon = \frac{J_t}{mL}. \tag{2}$$

Substituting Eq. (2) into the first of Eqs. (1) and using $\Pi_b\ddot{b} = \ddot{b} + \|\dot{b}\|^2 b$ shows that the point $y = r + \varepsilon b$, the center of

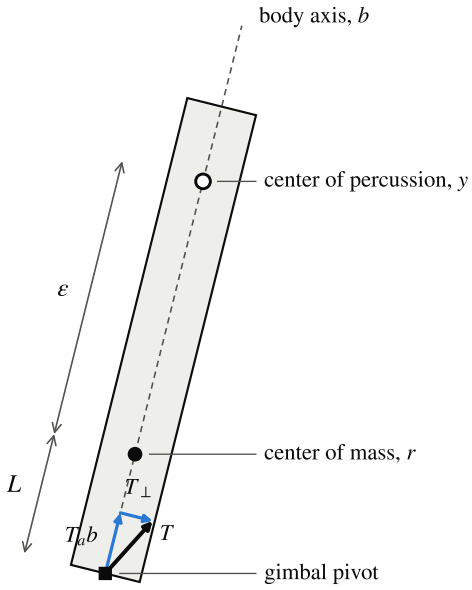


Fig. 1 Vehicle geometry. The gimbal angle is exaggerated.

percussion, obeys

$$\ddot{y} = \Big(\frac{T_a}{m} - \varepsilon\|\dot{b}\|^2\Big)b - g_0 e_z, \tag{3}$$

provided ε is constant. The lateral thrust does not appear. Seen from y, the gimbaled vehicle is a point mass whose thrust is locked to the body axis. We assume throughout that ε is constant along the flight. For constant mass this holds exactly; with depleting mass it requires J_t to scale with mL. The mass-property table of the vehicle's simulator gives ε = 1.352, 1.340, and 1.321 m at 79.9, 75, and 70.4 kg, so the assumption holds to 1% over a descent.

IV. Why an Exact Convex Formulation Does Not Exist

Let F be the set of initial positions and velocities (r_0, v_0) from which the vehicle can land within its limits at a given final time. A convex program that receives (r_0, v_0) as LCvx does, through affine constraints, has a feasible set that is convex jointly in the data and the variables. Its projection onto the data is then convex. If the program were both sound and complete, that projection would equal F, so F would be convex. For 3-DOF descent F is convex, because a thrust direction that can switch instantly can realize any average of two maneuvers [14]. For a body that turns at a finite rate it is not: with a minimum throttle, the only way to shed vertical impulse is to tilt, and tilting at the limit commits the vehicle to one direction, so a maneuver that tilts left and its mirror image are both flyable whereas their average is not.

A companion report [15] makes this precise. For a model with a rate-limited body axis and a gimbal cone, with altitude and glide-slope constraints, any final time, and upright landing, it proves that F is not convex, and the proof is checked by the Lean 4 proof assistant [16, 17]. For the gimbaled rigid body of Sec. 3 it gives computer-assisted proofs at final times of 1 to 5 s under simplifying assumptions (constant mass and an unlimited gimbal rate). We reproduced the obstruction numerically on the model and planning limits of this paper with a nonconvex solver [18]: on a 4-s hop launched vertically at 6.3 m/s, plans exist that land 1, 2, and 3 m from the launch point, but none was found that lands on it, from symmetric initial guesses or from nine asymmetric ones. The set of reachable landing points has a hole. This is numerical evidence, not a proof of infeasibility.

The conclusion is narrower than it may seem. Exactness, which is soundness and completeness together, is impossible. Soundness alone is not, and Eq. (3) makes it inexpensive.

V. A Sound Convex Program

A. Flat Map

Let $y(t)$ be four times continuously differentiable and define $u = \ddot{y} + g_0 e_z$ and $f = \|u\|$. Wherever $u \neq 0$, let

$$b = \frac{u}{f}, \qquad \dot{b} = \frac{\Pi_b \dot{u}}{f}, \qquad \Pi_b \ddot{b} = \frac{\Pi_b \ddot{u} - 2\langle b, \dot{u}\rangle \dot{b}}{f}, \qquad a = f + \varepsilon \|\dot{b}\|^2, \qquad \tau = \sqrt{a^2 + \varepsilon^2 \|\Pi_b \ddot{b}\|^2},$$
$$m(t) = m_0 \exp\Big(-\alpha \int_0^t \tau \, \mathrm{d}t\Big), \qquad T_a = ma, \qquad T_\perp = -m\varepsilon \, \Pi_b \ddot{b}, \qquad r = y - \varepsilon b. \tag{4}$$

The attitude, the body rate, the gimbal angle ($\tan\mu = \varepsilon\|\Pi_b\ddot{b}\|/a$), and the throttle ($\|T\| = m\tau$) are algebraic functions of y and its first four derivatives, and the mass is a quadrature. Section 6 shows that Eqs. (4) are the unique solution of Eqs. (1). Any sufficiently smooth curve y with $u \neq 0$ is therefore a flight of the vehicle, and only the actuator limits remain to be imposed.

B. Spline Parametrization

The unknown is y, a clamped quintic B-spline with S uniform spans of length $h = t_f/S$. It has $S + 5$ coefficients per axis, 63 scalars for $S = 16$, and it is four times continuously differentiable, so the gimbal angle and the throttle are continuous. On each span, u, $\dot{u}$, and $\ddot{u}$ are polynomials of degree 3, 2, and 1. Raising their Bézier control points to degree 3 gives four *control tuples* (U^j, U_1^j, U_2^j), $j = 0, \ldots, 3$, that are linear in the spline coefficients and satisfy

$$(u, \dot{u}, \ddot{u})(t) = \sum_{j=0}^{3} B_j(t) \, (U^j, U_1^j, U_2^j), \qquad B_j(t) \geq 0, \qquad \sum_{j=0}^{3} B_j(t) = 1, \qquad \int_{\text{span}} B_j \, \mathrm{d}t = \frac{h}{4}, \tag{5}$$

where B_j are the cubic Bernstein polynomials of the span. Equation (5) is what allows limits to be enforced at every instant and not only at grid points [12].

C. The Program

The data are the current position r_0, velocity v_0, body axis b_0, its rate $\dot{b}_0$, the lateral angular acceleration $\beta_0 = \Pi_b \ddot{b}(0)$ that the present gimbal deflection produces, the mass m_0, and the time of flight t_f; the target is the origin. Let $p = \langle U, e_z\rangle$ for a control tuple, $f_0 = \langle b_0, u(0)\rangle$, and $\Pi_0 = I - b_0 b_0^{\mathsf{T}}$. With auxiliary scalars q, k, G, and σ on each tuple and two constants $\phi > 0$ and $\ell_b > 0$, the program is

$$\begin{aligned}
\text{minimize} \quad & \tfrac{h}{4} \textstyle\sum \sigma && \text{(6a)}\\
\text{subject to} \quad & y(0) = r_0 + \varepsilon b_0, \quad \dot{y}(0) = v_0 + \varepsilon \dot{b}_0, \quad u(0) = f_0 b_0, \quad \Pi_0 \dot{u}(0) = f_0 \dot{b}_0 && \\
& \Pi_0 \ddot{u}(0) = f_0 \beta_0 + 2\langle b_0, \dot{u}(0)\rangle \dot{b}_0 && \text{(6b)}\\
& y(t_f) = \varepsilon e_z, \quad \dot{y}(t_f) = 0, \quad u_h(t_f) = \dot{u}_h(t_f) = \ddot{u}_h(t_f) = 0 && \text{(6c)}\\
& \|U_h\| \leq \tan\vartheta_{\max} \, p, \qquad p \geq \ell_b, \qquad \|U_1\|^2 \leq q\, p && \text{(6d)}\\
& \|U_2 - k e_z\| + \sin\vartheta_{\max} \, |k| + q \leq G, \qquad \varepsilon G \leq \tan\delta_{\max} \, (2\phi p - \phi^2) && \text{(6e)}\\
& \Big\|\big(\|U\| + \varepsilon q/\ell_b, \; \varepsilon G/\ell_b\big)\Big\| \leq \sigma && \text{(6f)}\\
& p \geq T_{\min} \, c_s(z_{lo,s}), \qquad \sigma \leq T_{\max} \, \tau_s(z_{up,s}) && \text{(6g)}
\end{aligned}$$

where the subscript h denotes the horizontal part of a vector, Eqs. (6d–6g) hold on every control tuple, and s indexes spans. The quantities in Eq. (6g) bracket the logarithm of the mass:

$$z_{up,s} = \ln m_0 - \alpha(\dot{y}_z(t_s) - \dot{y}_z(0) + g_0 t_s), \qquad z_{lo,s} = \ln m_0 - \alpha \, \tfrac{h}{4} \sum_{\text{spans} \leq s} \sum_j \sigma^j, \tag{7}$$

with t_s the start of span s. Both are affine in the unknowns. The function c_s is the chord of e^{-z} on an interval to which $z_{lo,s}$ is confined by two further linear constraints, and τ_s is a tangent of e^{-z}; the dry-mass limit is $z_{lo,S} \geq \ln m_{dry}$. Every constraint is linear or a second-order cone, so Eqs. (6) form an SOCP. The boundary conditions are linear because f_0 and $\langle b_0, \dot{u}(0)\rangle$ are linear in the coefficients, and (r_0, v_0) enter affinely, exactly as in LCvx.

The reading of the constraints is as follows. Equations (6b) start the plan at the present position, velocity, attitude, body rate, and gimbal deflection. Equations (6c) end it at rest on the target, upright, without rotation, and with the gimbal centered. The first of Eqs. (6d) is the tilt limit, and it is exact. The second of Eqs. (6e) is the gimbal limit: the snap and the squared jerk of the center-of-percussion path are bounded by the authority of the gimbal. Equations (6f) and (6g) are the throttle limits, with the vertical part of u standing in for its magnitude in the lower one. The altitude floor and a glide slope on r are added at sampled instants with a slack variable, as is usual. The constants are $\phi = 7.5$ m/s^2, the specific force at which the gimbal constraint is tightest, and $\ell_b = T_{\min}/m_0$. Any positive values keep the guarantee of Sec. 6; they change only how much of the feasible set is retained.

VI. Soundness

Theorem 1. *Assume ε is constant and $\angle(b_0, e_z) \le \vartheta_{\max}$, and let y be feasible for Eqs.* (6)*. Then Eqs.* (1)*, started from $(r_0, v_0, b_0, \dot b_0, m_0)$ and driven by the thrust $T = T_a b + T_\perp$ of Eqs.* (4)*, have the unique solution $r = y - \varepsilon b$, and this solution 1) begins with the lateral thrust the gimbal is producing, $T_\perp(0) = -m_0\varepsilon\beta_0$; 2) ends with $r = \dot r = 0$, $b = e_z$, $\dot b = 0$, and $T_\perp = 0$; 3) satisfies $T_{\min} \le \|T\| \le T_{\max}$, $\mu \le \delta_{\max}$, and $\angle(b, e_z) \le \vartheta_{\max}$ at every $t \in [0, t_f]$; and 4) consumes at most $m_0[1 - \exp(-\alpha\frac{h}{4}\sum\sigma)]$ of propellant.*

Proof. Step 1: the flat map solves the equations of motion. Differentiating $u = fb$ twice and projecting with Π_b gives $\Pi_b\dot u = f\dot b$ and $f\,\Pi_b\ddot b = \Pi_b\ddot u - 2\dot f\dot b$ with $\dot f = \langle b, \dot u\rangle$, so the quantities named $\dot b$ and $\Pi_b\ddot b$ in Eqs. (4) are the derivatives of b. Take $\Omega = b\times\dot b$. Then $\Omega \perp b$, $\Omega\times b = \dot b$, and $\dot\Omega = b\times\Pi_b\ddot b$. The torque of the thrust is $(-Lb)\times T = Lm\varepsilon\, b\times\Pi_b\ddot b = J_t\dot\Omega$, because $m\varepsilon L = J_t$. The force is $T - mg_0e_z = m(f + \varepsilon\|\dot b\|^2)b - m\varepsilon(\ddot b + \|\dot b\|^2 b) - mg_0e_z = m(\ddot y - \varepsilon\ddot b) = m\ddot r$. Finally $\|T\| = m\tau$ and $\dot m = -\alpha\tau m$ by the definition of m. The right-hand sides of Eqs. (1) are locally Lipschitz in the state, so the solution is unique.

Step 2: boundary conditions. The end control points of a Bézier polygon are values of the curve, so $u_z \ge \ell_b > 0$ at $t = 0$ and $t = t_f$. At $t = 0$, $u(0) = f_0b_0$ with positive vertical components forces $f_0 > 0$ and $b(0) = b_0$; the next two conditions of Eq. (6b) then give $\dot b(0) = \dot b_0$ and $\Pi_b\ddot b(0) = \beta_0$ by Step 1, and the first two give $r(0) = r_0$ and $\dot r(0) = v_0$. At t_f, $u_h = 0$ gives $b = e_z$, then $\dot u_h = 0$ gives $\dot b = 0$, then $\ddot u_h = 0$ gives $\Pi_b\ddot b = 0$, and $r = y - \varepsilon e_z = 0$, $\dot r = \dot y = 0$.

Step 3: pointwise bounds. Fix an instant and suppose that numbers q, k, G, σ satisfy Eqs. (6d–6f) with (U, U_1, U_2) replaced by $(u, \dot u, \ddot u)$. Let $p = u_z \le f$. The first of Eqs. (6d) is $\tan\angle(b, e_z) \le \tan\vartheta_{\max}$. Next, $f\|\dot b\|^2 = \|\Pi_b\dot u\|^2/f \le \|\dot u\|^2/p \le q$. For the angular acceleration, $f\,\Pi_b\ddot b = \Pi_b(\ddot u - ke_z) + k\,\Pi_be_z - 2\langle b, \dot u\rangle\Pi_b\dot u/f$. The norm of the first term is at most $\|\ddot u - ke_z\|$; $\|\Pi_be_z\| = \sin\angle(b, e_z) \le \sin\vartheta_{\max}$; and $2|\langle b, \dot u\rangle|\|\Pi_b\dot u\| \le \|\dot u\|^2$ because the two factors are the parts of $\dot u$ along and across b. Hence $f\|\Pi_b\ddot b\| \le G$. Since $f^2 \ge p^2 \ge 2\phi p - \phi^2$, the second of Eqs. (6e) gives $\varepsilon\|\Pi_b\ddot b\| \le \tan\delta_{\max} f \le \tan\delta_{\max}\, a$, which is $\mu \le \delta_{\max}$. Lastly $p \le f \le a \le \tau$, and $a \le \|u\| + \varepsilon q/\ell_b$ and $\varepsilon\|\Pi_b\ddot b\| \le \varepsilon G/\ell_b$ because $f \ge p \ge \ell_b$, so $p \le \tau \le \sigma$ by Eq. (6f).

Step 4: every instant. Equations (6d–6f) describe a convex set of points $(U, U_1, U_2, q, k, G, \sigma)$ that is the same for the four control tuples of a span. By Eq. (5), $(u, \dot u, \ddot u)(t)$ together with the Bernstein-weighted auxiliary numbers is a convex combination of four points of that set, so it lies in the set. Step 3 therefore applies at every instant, with $\sigma(t) = \sum_j B_j(t)\sigma^j$.

Step 5: mass. Let $z = \ln m$, so that $\dot z = -\alpha\tau$ and $p \le \tau \le \sigma(t)$. On span s, $z(t) \le z(t_s) \le \ln m_0 - \alpha\int_0^{t_s} u_z\,\mathrm{d}t = z_{up,s}$ and $z(t) \ge z(t_{s+1}) \ge \ln m_0 - \alpha\int_0^{t_{s+1}}\sigma\,\mathrm{d}t = z_{lo,s}$, using the last of Eqs. (5). Because e^{-z} is decreasing and convex, a chord lies above it and a tangent below it, so Eq. (6g) gives $p(t) \ge T_{\min}e^{-z(t)}$ and $\sigma(t) \le T_{\max}e^{-z(t)}$, that is, $T_{\min} \le mp \le m\tau = \|T\| \le m\sigma \le T_{\max}$. The fuel bound is $z(t_f) \ge z_{lo,S}$. □

No relaxation is used, so there is no tightness to establish; this is the difference from LCvx. The price is stated next.

Proposition 1. *For fixed b_0, $\dot b_0$, β_0, m_0, and t_f, the set of (r_0, v_0) at which Eqs.* (6) *are feasible is convex and is contained in F. Whenever F is not convex, the containment is strict.*

Proof. The data enter only the right-hand sides of the first two conditions of Eq. (6b), so the projection argument of Sec. 4 gives convexity. Containment is Theorem 1. A convex set cannot equal a nonconvex one. □

The program must therefore refuse some landings that exist. The loss has three sources: the lower throttle limit is imposed on the vertical part of the thrust, so the program never plans to shed thrust by tilting, which is the maneuver the obstruction of Sec. 4 is built from; the interaction of throttle changes with rotation is bounded, not computed; and control polygons contain the curves they bound. Section 8 measures the size of the loss. Two limits are outside the

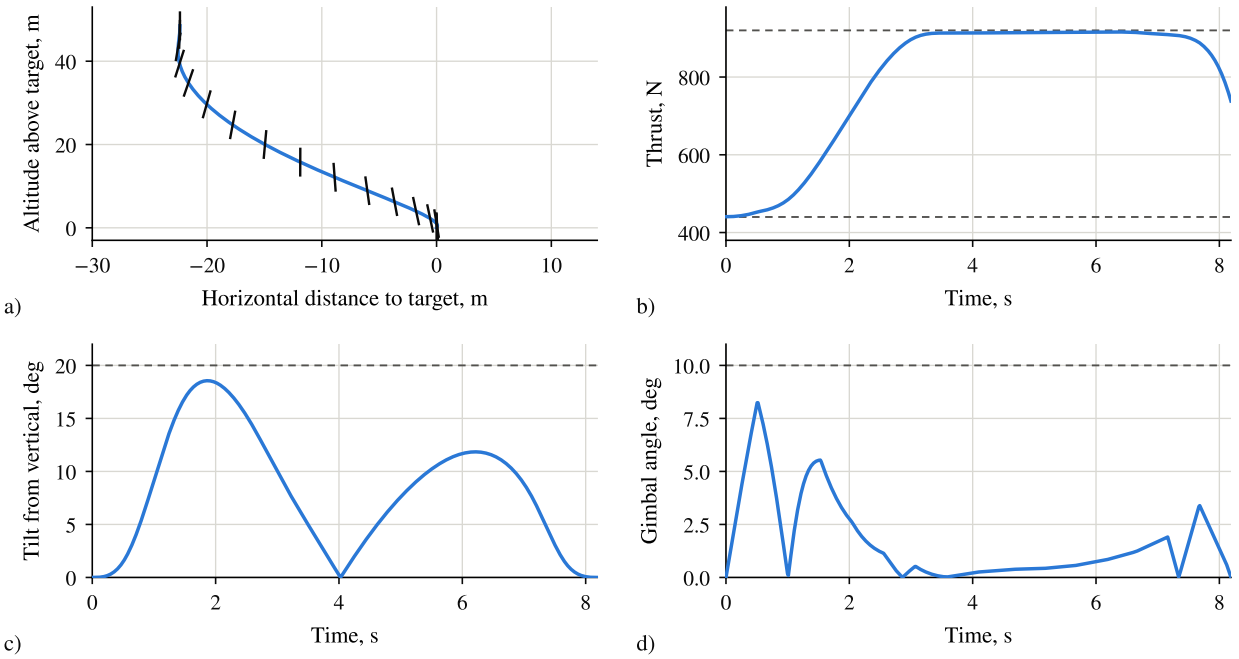


Fig. 2 One plan from one solve of Eqs. (6)**, a descent from 50 m altitude and 22 m offset: a) path of the center of mass, with the body axis drawn every 0.5 s; b) thrust; c) tilt; and d) gimbal angle. Dashed lines are the planning limits.**

theorem. The gimbal and throttle slew rates depend on the fifth derivative of y and are not constrained by Eqs. (6); they are evaluated exactly on every plan, and a plan that violates them is not flown. The altitude and glide-slope constraints hold at the sampled instants only.

VII. Guidance Algorithm

The guidance law has three steps. 1) Once, before the descent, the time of flight is set to the smallest t_f at which Eqs. (6) are feasible, found by bisection in about ten solves, plus a margin of 0.25 s. 2) At every guidance step the program is solved for the remaining time of flight from the current state estimate. The same program is solved each time; nothing is carried over from the previous plan, and no initial trajectory, trust region, or penalty weight is required. 3) At every control step the reference position, velocity, attitude, body rate, throttle, and gimbal angle are evaluated from Eqs. (4) in closed form and passed to the tracking controller. If a solve is infeasible or its plan fails the slew-rate check, the final condition $u_z(t_f) = g_0$ (arrival in hover, used by default to soften the touchdown) is dropped, then the flight time is extended by 0.25, 0.5, or 1.0 s, and otherwise the previous plan, which remains valid, is kept. Figure 2 shows one plan: the vehicle tilts toward the target, tilts back to brake, and arrives upright, with the thrust rising smoothly from its lower to its upper planning limit.

For $S = 16$ the program has 63 spline coefficients. As passed to the solver, with its auxiliary variables, it has about 1000 variables and 480 second-order cones, and one solve takes about 40 ms. The cones are of the types used by 3-DOF LCvx, so an existing LCvx flight implementation can host the program by changing the constraint builder only; the spline matrices depend only on S and are computed once.

VIII. Results

A. Setup

All results are numerical and use the vehicle of Table 1. The baseline, denoted S, is a standard 6-DOF SCvx implementation [5, 6] on the quaternion rigid body with 35 intervals, virtual control, and a trust region; it uses the same planning limits. The program of Eqs. (6) is denoted F and uses $S = 16$. Both are solved with the Clarabel interior-point solver, and solver times are those it reports on a four-core desktop processor. The simulated plant is the quaternion rigid

Table 2 Single plans on eight in-air scenarios

	S (successive convexification)	F (this work)
Convex programs per plan	25 to 35	1
Solver time per plan, median, s	1.30	0.046
Fuel at equal final time, relative to S	reference	+0.10% mean, +0.28% maximum
Minimum feasible final time, s	reference[a]	+0.04 to +0.22
Open-loop replay error, planner's own model	2 nm to 10 μm	at most 23 nm
Open-loop replay error, plant mass table, median, m	0.45	0.03

[a]Minimum certified final time of the iterative planner of Ref. [15], at which S also converges.

body with mass-dependent inertia and lever arm, a ±15 deg gimbal box with a 100 deg/s slew limit, a 400 to 1000 N throttle with a 2000 N/s slew limit, wind of up to 3 m/s with gusts, 2% thrust noise, a 5% error in specific impulse, and gimbal misalignment. One tracking controller, a linear-quadratic regulator in center-of-percussion coordinates running at 50 Hz, is used for both planners. A landing is counted as successful if the touchdown speed is at most 1 m/s, the lateral error at most 1 m, and the tilt at most 5 deg. The test set is 40 in-air descents (initial altitude 40 to 60 m, offset up to 21 m, initial tilt up to 3 deg) and 30 hops launched from the pad at 5 to 6 m/s toward a point up to 4 m away. The final time for S is 0.5 s above the minimum final time found for an iterative planner of the companion report [15]. For reference, the 3-DOF LCvx guidance currently used on the vehicle, which plans thrust directions the body cannot follow, landed 11 of the 40 in-air cases and none of the hops in that report.

B. Open-Loop Comparison

Table 2 compares single plans on eight in-air scenarios at equal final time. At a fixed final time the fuel is nearly the same for any planner, because the boundary conditions fix the total impulse; F uses 0.10% more on average. The difference is the work: one SOCP against 25 to 35. Program (6) needs a slightly longer flight at the limit, which is the incompleteness of Proposition 1.

C. Exactness, Model Error, and Soundness

To test Step 1 of the proof, the throttle and gimbal commands of F were applied without feedback to the quaternion rigid body with depleting mass and constant ε, integrated to a relative tolerance of 10^{-13}. In 16 replays, half of them starting tilted by 3 deg and rotating, the vehicle came to rest within 23 nm and 12 nm/s of the target, which is integration error. Each planner is exact in its own model. S freezes J_t and L at their initial values, which lets $J_t/(mL)$ drift by 6% over a descent, whereas the plant keeps it within 1%. Replayed without feedback against the plant's mass table, plans of S miss by a median of 0.45 m and plans of F by 0.03 m.

To test Theorem 1, 300 random problems, wider than the test set, were each solved with the constants above and with randomly chosen ones. Of the 600 solves, 458 were feasible, and none of the plans violated a planning limit at any of 1201 instants. To test Proposition 1, both planners and a nonconvex solver [18] were run on a grid of 20 hops of exactly 4 s (launch speeds of 5 to 7 m/s, landing offsets of 0 to 6 m). No flyable plan was found for 11 of the targets, and F is infeasible at all 11. It is also infeasible at five of the nine targets where a plan exists, which it reaches with 0.06 to 0.41 s more flight time. A 3-DOF LCvx program reports an optimal solution at all 20 targets.

D. Closed Loop with Computation Time Counted

Simulation studies of guidance commonly apply each plan at the instant its state was sampled. Here a plan computed from the state at time t becomes active at t plus its solver time, and each planner replans as soon as it can: S at most every 0.5 s and F every 0.1 s. F chooses its own time of flight as in Sec. 7. Table 3 and Fig. 3 give the results. Both planners land every case. F is more accurate laterally by more than an order of magnitude, touches down more slowly, and moves the gimbal a third as far. On hops it uses 5% more fuel, because it flies them 0.1 s longer on average and extended the flight during a replan in six runs. With every solver time multiplied by four, to represent a slower flight computer, S fails four of the 30 hops, and F, replanning every 0.2 s, fails none.

The source of the advantage should be stated carefully. When both planners replan at 1 Hz, plans are applied instantly, and both fly the same final time, the lateral accuracy is the same: a median of 3.1 cm for each in the air and 2.4 cm for each on hops. F then still touches down more slowly on hops (0.23 against 0.51 m/s) and never needs to extend the final time in the air, which S does in 32 of 40 runs. The accuracy in Table 3 therefore comes from the replanning rate, which a solver time of 30 ms permits and a solver time of 1 s does not. Giving S the arrival-in-hover

Table 3 Closed-loop results with each plan applied after its solver time (median / 95th percentile / maximum)

	In-air descents (40 runs)		Pad hops (30 runs)	
	S	F	S	F
Successful landings	40	40	30	30
Lateral error, cm	5.9 / 15.7 / 18.3	0.2 / 0.8 / 1.1	3.9 / 7.6 / 8.2	0.4 / 1.4 / 2.0
Touchdown speed, m/s	0.23 / 0.46 / 0.58	0.19 / 0.28 / 0.29	0.54 / 0.81 / 0.88	0.14 / 0.22 / 0.23
Tilt at touchdown, deg	0.21 / 0.70 / 1.23	0.11 / 0.30 / 0.63	0.96 / 2.46 / 2.54	0.20 / 0.62 / 0.87
Fuel, mean, kg	3.58	3.52	1.46	1.54
Solver time per replan, s (mean / maximum)	1.05 / 4.9	0.031 / 0.39	0.35 / 2.4	0.017 / 0.052
Convex programs per plan, mean	20.2	1.4	12.7	1.1
Gimbal travel per flight, mean, deg	156	45	182	62

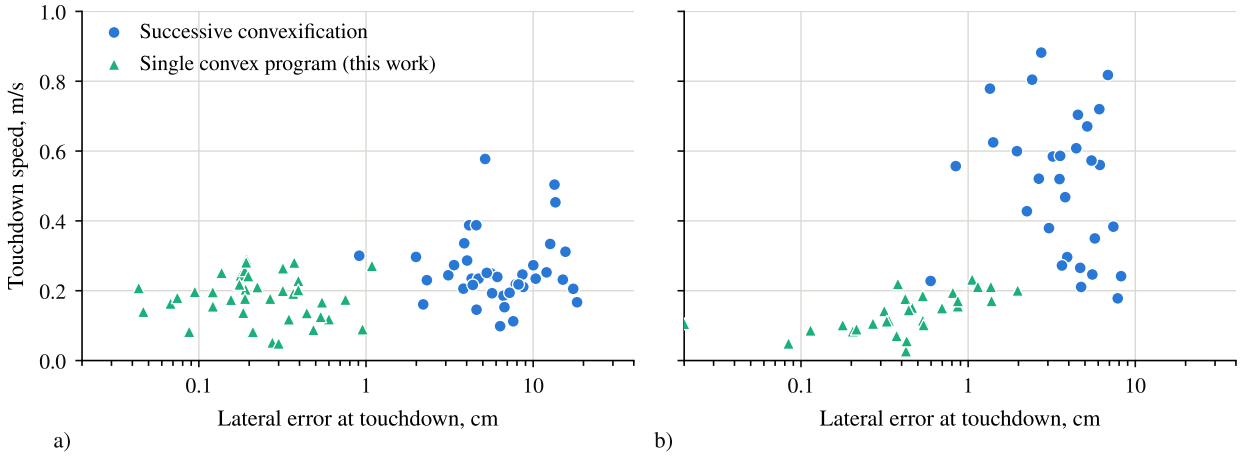


Fig. 3 Touchdown accuracy with computation time counted: a) 40 in-air descents and b) 30 pad hops.

condition does not change this (hops: 0.52 m/s and 4.0 cm).

E. Limitations

F is worse than S in three respects: fuel on short hops, the minimum flight time (by up to 0.4 s on 4-s hops), and, when S is given zero computation time, the touchdown speed in the air (S is lower in 27 of 40 cases). The slew-rate limits are outside Theorem 1; fewer than 1% of replans failed the check and were not flown. The worst replan took 0.39 s, when every fallback was tried, and a flight implementation should bound that chain. Theorem 1 is proved here and tested numerically but is not machine-checked, apart from Eq. (3), which is machine-checked for constant mass [15]. The assumption of constant ε must be confirmed from measured mass properties. Nothing here has been run on flight hardware, with actuator lag, or with state-estimation error, and the SCvx baseline was not retuned or reimplemented for speed.

IX. Conclusion

For a lander steered by one gimbaled engine, a convex program cannot accept exactly the landings that are flyable, but it can accept only flyable ones. Writing the problem for the center of percussion removes the coupling between steering and translation, makes the dynamics linear, and leaves the actuator limits as the only nonconvex part; restricting those limits, instead of relaxing them, yields a single second-order cone program whose plans are flyable at every instant by proof. The program costs a tenth of a percent of fuel and a few tenths of a second of minimum flight time. Its practical value is its speed and its lack of tuning: it has no initial guess, trust region, or iteration count, and it solves fast enough to replan at the rate of the control system, which in our simulations mattered more for touchdown accuracy than the last fraction of a percent of optimality. Future work is to bring the slew-rate limits inside the guarantee, to bound the loss of feasible landings, to formalize the proof, and to fly it.

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